

Task 6: Automated Cost Optimizer

Purpose

Reduce AWS bill automatically

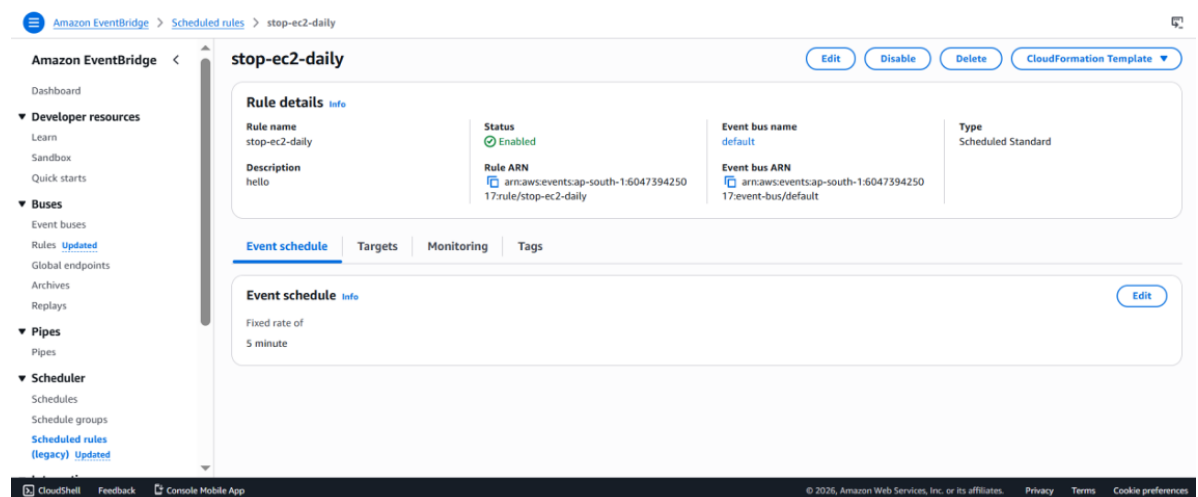
AWS Services

AWS Lambda

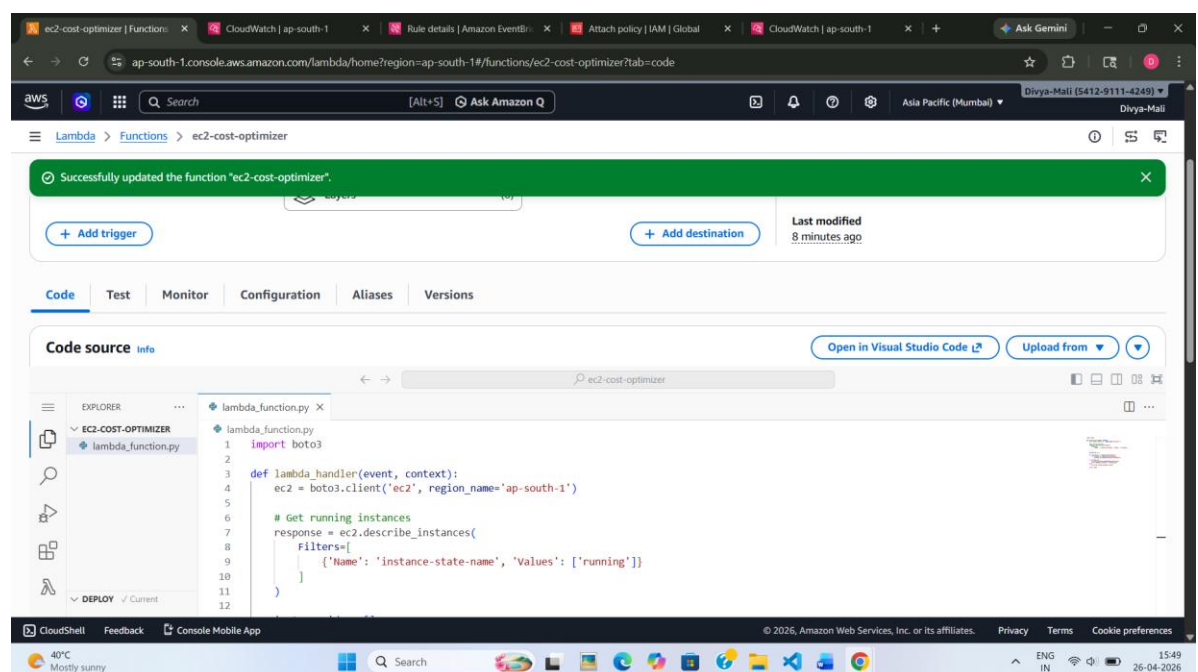
CloudWatch

EC2

Step 1:



The screenshot shows the Amazon EventBridge console for a rule named 'stop-ec2-daily'. The rule is enabled and has a description of 'hello'. It is configured with the 'arnaws:events:ap-south-1:6047394250:rule/stop-ec2-daily' Rule ARN and the 'arnaws:events:ap-south-1:6047394250:17:event-bus/default' Event bus ARN. The event schedule is set to a fixed rate of 5 minutes. The console includes a sidebar with navigation links for Dashboard, Developer resources, Buses, Pipes, and Scheduler. The top right has buttons for Edit, Disable, Delete, and CloudFormation Template.



The screenshot shows the AWS Lambda console for the 'ec2-cost-optimizer' function. A green notification bar at the top indicates 'Successfully updated the function "ec2-cost-optimizer"'. The function is currently in the 'Code' tab, showing the following Python code:

```
1 import boto3
2
3 def lambda_handler(event, context):
4     ec2 = boto3.client('ec2', region_name='ap-south-1')
5
6     # Get running instances
7     response = ec2.describe_instances(
8         Filters=[
9             {'Name': 'instance-state-name', 'Values': ['running']}
10        ]
11    )
12
```

The console also shows tabs for Test, Monitor, Configuration, Aliases, and Versions. The bottom of the screen displays the Windows taskbar with the date 26-04-2026 and time 15:49.

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms 1 1 0

▼ AI Operations

Overview

Investigations

Configuration

► GenAI Observability

▼ Application Signals (APM)

Services

Application Map

Transaction Search

Service Level Objectives

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns.](#)

Filter events - press enter to search

Clear 1m 30m 1h 12h Custom UTC timezone Display

Timestamp	Message
No older events at this moment. Retry	
2026-04-22T19:52:02.665Z	INIT_START Runtime Version: python:3.11.mainlinev2.v7 Runtime Version ARN: arn:aws:lambda:ap-south-1::runtime:c51a1796965c4cd7cb4442b973d67819a79e317f...
2026-04-22T19:52:03.173Z	START RequestId: 57f72eda-3e62-41f6-9126-15abf20525ab Version: \$LATEST
2026-04-22T19:52:03.670Z	Stopped Instance: 1-01a3d5bfc527ccbf
2026-04-22T19:52:03.677Z	END RequestId: 57f72eda-3e62-41f6-9126-15abf20525ab
2026-04-22T19:52:03.677Z	REPORT RequestId: 57f72eda-3e62-41f6-9126-15abf20525ab Duration: 497.84 ms Billed Duration: 1004 ms Memory Size: 128 MB Max Memory Used: 99 MB Init Du...
No newer events at this moment. Auto retry paused. Resume	